Momentum analysis: put it together

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a model to compare a deflection option and its limits.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Communicate assumptions with the numerical result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to compare your model or result. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

Total momentum initially is (15, 0) kg·m/s. One fragment leaves with (3, 5) kg·m/s.
Find the other momentum vector and magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: shift = Δv × time; J = MΔv.
Fictional straight-line planning model. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A 1 kg cart at +6 m/s sticks to a 2 kg cart at rest. Final velocity is +2 m/s. The moving cart changes momentum by -4 and the other by +4 kg·m/s.

Your turn: Draw both momentum changes and show how their sum matches the system change.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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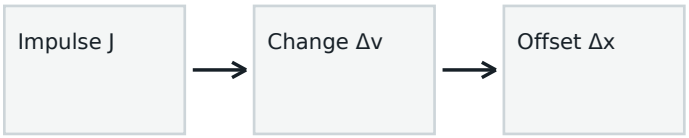
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

04 Apply the physics

Target mass 2×10^{9} kg; desired shift 20 km over 4×10^{6} s.
Find required Δv and impulse.
Compare an option delivering 8×10^{6} N·s. Organize the known quantities on the before-and-after model.

Sentence starter: Both … . They differ in … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

If the target mass estimate is too low, how would a fixed delivered impulse's velocity change compare with the prediction?

Word help: estimate = find an approximate value

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg cart moving right at 6 m/s stops in 3 s. Take right as positive.
Find signed impulse and average force.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, including direction

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Keep impulse fixed but double both target mass and lead time. What happens to modeled shift? Justify with an equation. Organize the known quantities on the before-and-after model.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Under the assumption …, the evidence … supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart at +4 m/s and a 1 kg cart at -2 m/s stick while an external impulse of -3 N·s acts.
Find final velocity and explain why setting initial momentum equal to final would fail. Organize the known quantities on the before-and-after model.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3